Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Science, Technology, Engineering and Mathematics

School of Mathematics

SS Mathematics Semester 2 **2024**
SS Theoretical Physics

MAU44400 Quantum Field Theory

30/04/2024 RDS - MAIN 14.00 - 17.00

Prof Jan Manschot and Prof Samson Shatashvili

Instructions to candidates:

Credit will be given to the best two questions from Part A (Questions 1, 2 and 3) and the best two questions from Part B (Questions 4, 5 and 6).

Please complete Part A and Part B on separate answer booklets.

'Formulae and Tables' are available from the invigilators, if required.

Non-programmable calculators are permitted for this examination.

You may not start this examination until you are instructed to do so by the Invigilator.

1. Massive gauge field

a) **10 marks**: Derive the equations of motion for the following Action functional:

$$
\begin{aligned}
A(\phi^\mu, \partial_\nu \phi^\mu) &= \int d^4x \left[-\frac{1}{2}(\partial_\mu \phi^\nu)^2 + \frac{1}{2}(\partial_\mu \phi^\mu)^2 + \frac{m^2}{2}\phi_\mu \phi^\mu \right] \\
&= \int d^4x \left[-\frac{1}{4}(\partial_\mu \phi_\nu - \partial_\nu \phi_\mu)(\partial^\mu \phi^\nu - \partial^\nu \phi^\mu) + \frac{m^2}{2}\phi_\mu \phi^\mu \right]
\end{aligned}
$$

b) **5 marks**: Show that these equations imply $\partial_\mu \phi^\mu = 0$.

c) **10 marks**: Construct the Hamiltonian density, and constraints if there are such, as functionals of $\phi_\mu, \partial_i \phi_\mu$ and generalized momentum Π_μ.

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2. Maxwell-Chern-Simons theory

Consider the following action in three space-time dimensions

$$S_{CS} = k \int d^3x \mathcal{L}_{CS} = k \int d^3x \, \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho.$$

It is known as the Chern-Simons action.

a) **10 marks.** Demonstrate that it is gauge invariant. Show that any solution to the equations of motion for the above action is gauge-equivalent to a trivial solution $A_\mu = 0$.

b) **15 marks.** Consider the following, Maxwell-Chern-Simons, action:

$$S = -\frac{1}{4e^2} \int d^3x F_{\mu\nu} F^{\mu\nu} + k \int d^3x \, \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho.$$

Find the Hamiltonian.

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3. Current-current commutators in 2 space-time dimensions

Consider left-handed fermions in two space-time dimensions (t, x): $\psi_L = \frac{1}{2}(1-\gamma_5)\psi_D$:

Define the current:

$$J_0^\epsilon(t,x) = \bar{\psi}_L(t, x+\epsilon)\gamma^0\psi_L(t, x-\epsilon) = \psi_L^+(x+\epsilon)\psi_L(x-\epsilon)$$

a) **15 marks:** Use canonical equal time (anti)commutation relations for fermions $\{\psi_{D,a}(x), \psi_{D,b}^+(y)\} = \delta_{ab}\delta(x-y)$ and compute equal time commutators for these currents:

$$[J_0^\epsilon(t,x), J_0^\epsilon(t,y)]$$

b) **10 marks:** For the massless case:

$$< 0|\psi_L^+(t,x)\psi_L(t,y)|0 >= \frac{1}{x-y}$$

Evaluate

$$< 0|[J_0^\epsilon(t,x), J_0^\epsilon(t,y)]|0 >$$

and its limit for $\epsilon \to 0$.

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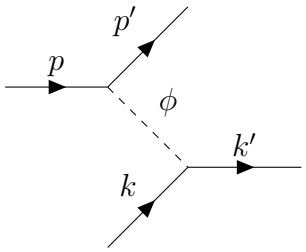
4. Yukawa theory

Consider the Yukawa theory with interaction Hamiltonian

$$H_I = g \int d^3\vec{x}\, \bar{\psi}\psi\phi,$$

and masses m_ϕ of the scalar field and m of the electron.

a) **10 marks:** Consider the Feynman diagram



Explain to which Wick contractions of the expansion of

$$\left\langle \vec{p}', \vec{k}' \left| e^{-i\int_{-T}^{T} H_I dt} \right| \vec{p}, \vec{k} \right\rangle$$

this amplitude corresponds, and determine the amplitude $\mathcal{M}(p, k \to p', k')$.

b) **5 marks:** Derive that the amplitude reduces to

$$i\mathcal{M}((p,s),(k,r) \to (p',s'),(k',r')) = \frac{4i\, g^2\, m^2}{\left|\vec{p}' - \vec{p}\right|^2 + m_\phi^2} \delta^{ss'}\delta^{rr'},$$

in the non-relativistic limit. Here s, s', r, r' label the components of the spinors.

c) **10 marks:** Derive the Yukawa potential

$$V(r) = -\frac{g^2}{4\pi}\frac{1}{r}e^{-m_\phi r}.$$

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5. **Compton scattering**

Consider Compton scattering $e^-\gamma \to e^-\gamma$.

a) **10 marks:** Give the two tree level Feynman diagrams which contribute $O(\alpha) = O(e^2)$ to this process. Use p (p') for the incoming (outgoing) electron, and k (k') for the incoming (outgoing) photon.

b) **5 marks:** Let $\mathcal{M}_1$ and $\mathcal{M}_2$ be the two amplitudes corresponding to the two diagrams. We can express these as $\mathcal{M}_j = \epsilon^*_\mu(k')\mathcal{M}^\mu_j$, $j = 1, 2$. Explain which diagram of your answer under part $a)$ corresponds to

$$\mathcal{M}^\mu_1 = e^2\epsilon_\nu(k)\bar{u}(p')\left[\frac{-\gamma^\nu \not{k}'\gamma^\mu + 2\gamma^\nu p^\mu}{2p\cdot k'}\right]u(p)$$

c) **10 marks:** Evaluate the scalar product of $(k')_\mu\mathcal{M}^\mu_1$. Argue (without explicit calculation) what $(k')_\mu\mathcal{M}^\mu_2$ evaluates to.

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6. Radiative corrections

a) **10 marks:** Consider radiative corrections to electron-to-electron scattering. In terms of the electron-to-electron cross-section $d\sigma(p \to p')$, the differential cross-section for the emission of a photon with 4-momentum k reads,

$$d\sigma(p \to p' + k_\gamma) = d\sigma(p \to p')\, \frac{\alpha}{2\pi}\, \log\left(-\frac{q^2}{\mu^2}\right) \mathcal{I}(p, p'),$$

with $q = p - p'$, μ a fictitious mass to regulate the infrared divergence, and

$$\mathcal{I}(p, p') = \int \frac{d\Omega_{\vec{k}}}{4\pi} \left(\frac{2p \cdot p'}{(\hat{k} \cdot p)(\hat{k} \cdot p')} - \frac{m^2}{(\hat{k} \cdot p)^2} - \frac{m^2}{(\hat{k} \cdot p')^2} \right)$$

where $\hat{k} = (1, \vec{k})$ and $\hat{k}^2 = 0$. Derive

$$\mathcal{I}(p, p') = -2 + \int_0^1 \frac{2m^2 - q^2}{m^2 - x(1-x)q^2}\, dx,$$

and the leading behavior for $-q^2 \to \infty$.

b) **5 marks:** Argue that radiative corrections to the electron vertex $\Gamma^\mu(p, p')$ must take the general form

$$\Gamma^\mu(p', p) = \gamma^\mu\, F_1(q^2) + \frac{i\sigma^{\mu\nu}}{2m} F_2(q^2)$$

with F_1 and F_2 the form factors.

c) **10 marks:** Recall that the first terms of the α-expansion of $F_1(q^2)$ are

$$F_1(q^2) = 1 - \frac{\alpha}{2\pi} f_{\mathrm{IR}}(q^2) \log\left(-\frac{q^2}{\mu^2}\right) + O(\alpha^2)$$

for a specific function $f_{\mathrm{IR}}(q^2)$ and μ the infrared cutoff as in Part a). Argue that the physically measured cross-section is independent of μ.

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